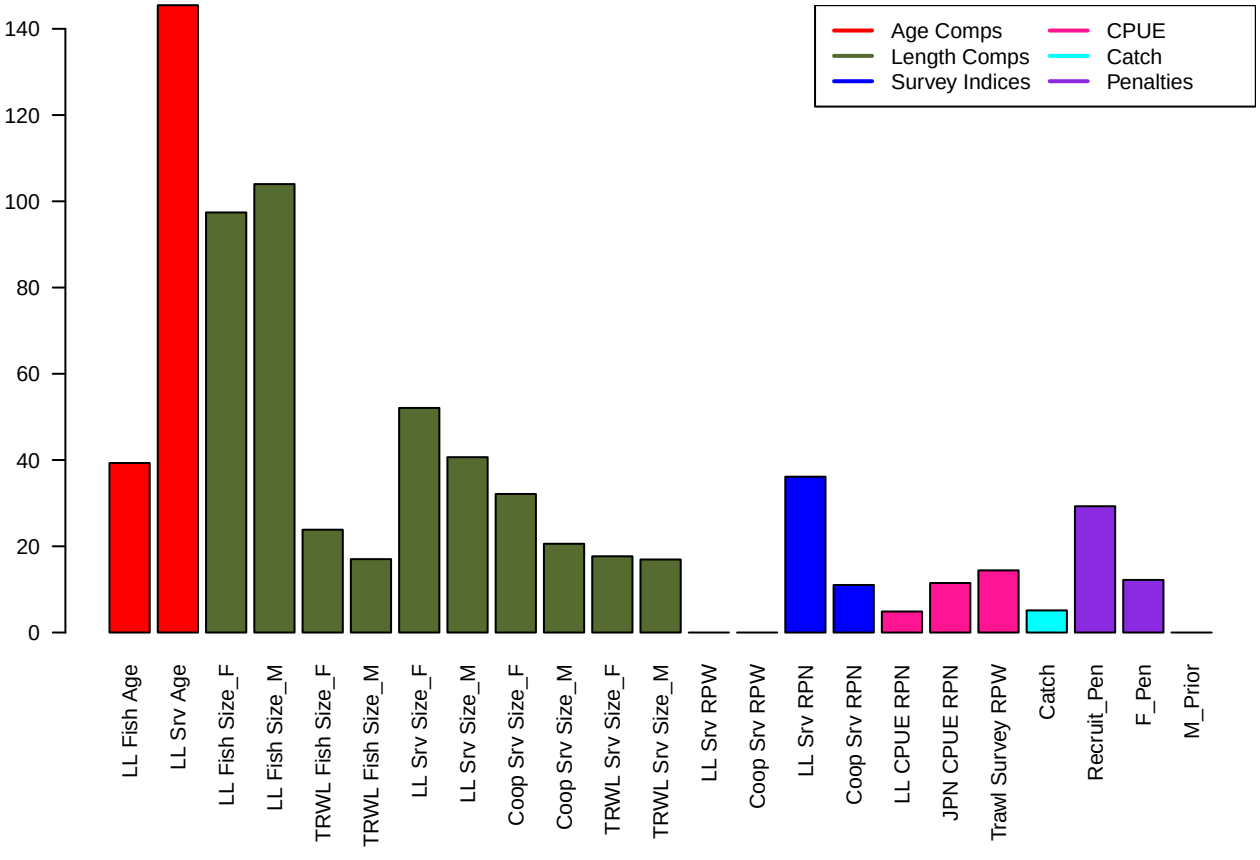
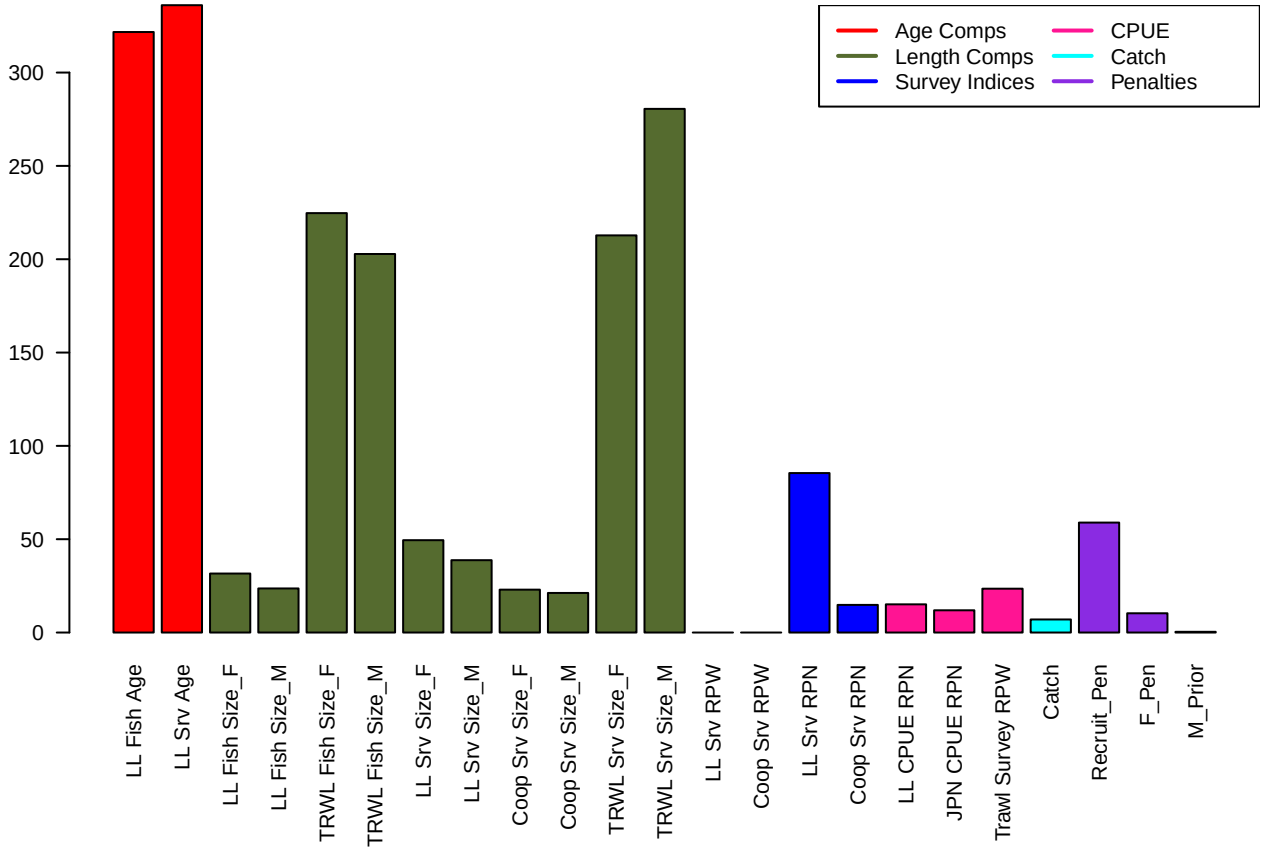


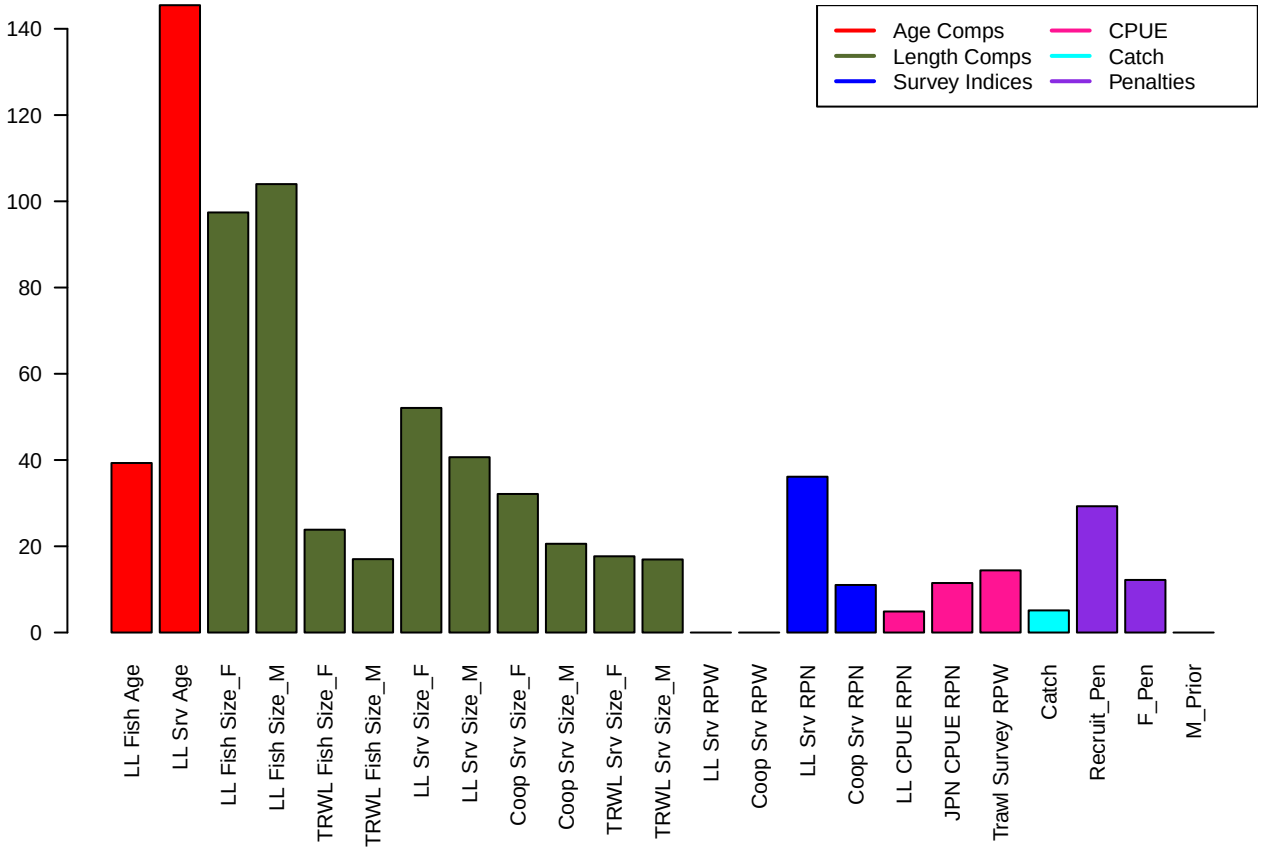
Model	Converged?	Max Grad	nLL	# Pars
21.12_Proposed_No_Skip_Spawn	TRUE	0.000678008981443245	752.92	252

2021 SSB (kt)	SSB_40 (kt)	2021 SSB/SSB_40	Mean_Recruit	2021 F	F_40	2021 F/F_40	F_ABC	2022 ABC (kt)
107.47	118.14	0.91	20.37	0.05	0.08	0.62	0.08	34.84



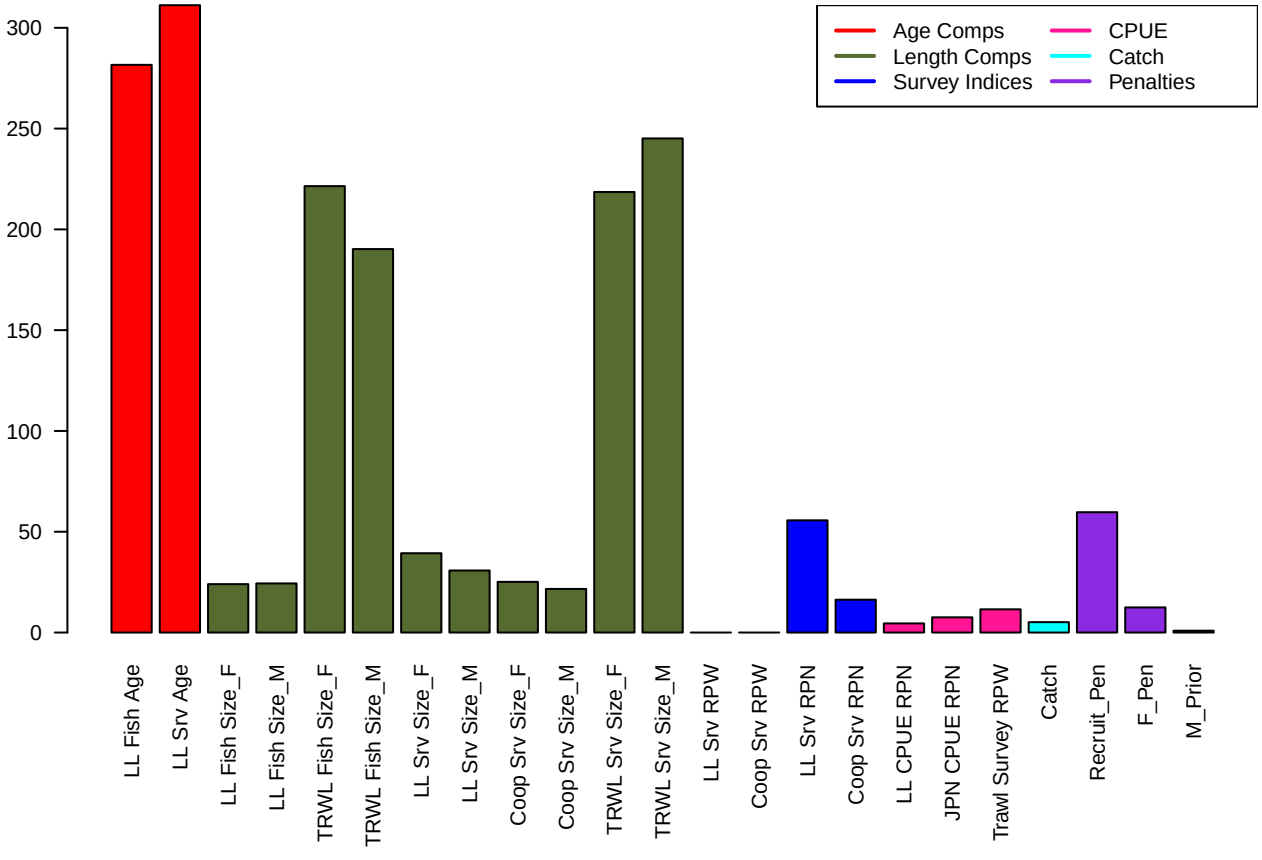
Model	Converged?	Max Grad	nLL	# Pars
16.5_Cont	TRUE	0.000205493656629296	2138.41	243

2021 SSB (kt)	SSB_40 (kt)	2021 SSB/SSB_40	Mean_Recruit	2021 F	F_40	2021 F/F_40	F_ABC	2022 ABC (kt)
133.64	126.63	1.06	24.37	0.04	0.1	0.4	0.1	67.43



Model	Converged?	Max Grad	nLL	# Pars
21.10_Proposed_w_Skip_Spawn	TRUE	0.000977841500663129	752.91	252

2021 SSB (kt)	SSB_40 (kt)	2021 SSB/SSB_40	Mean_Recruit	2021 F	F_40	2021 F/F_40	F_ABC	2022 ABC (kt)
102.18	116.54	0.88	20.37	0.05	0.08	0.62	0.08	34.65

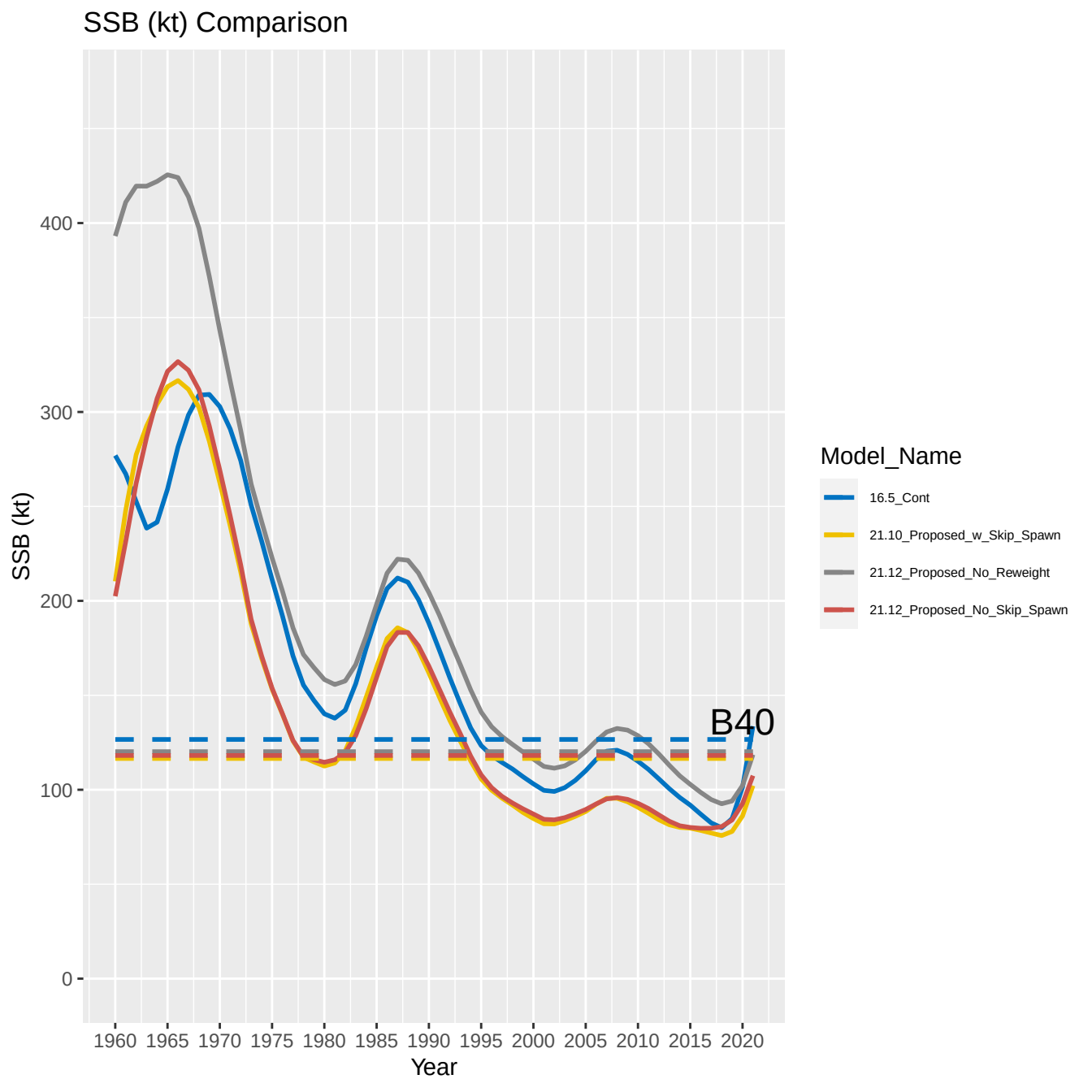


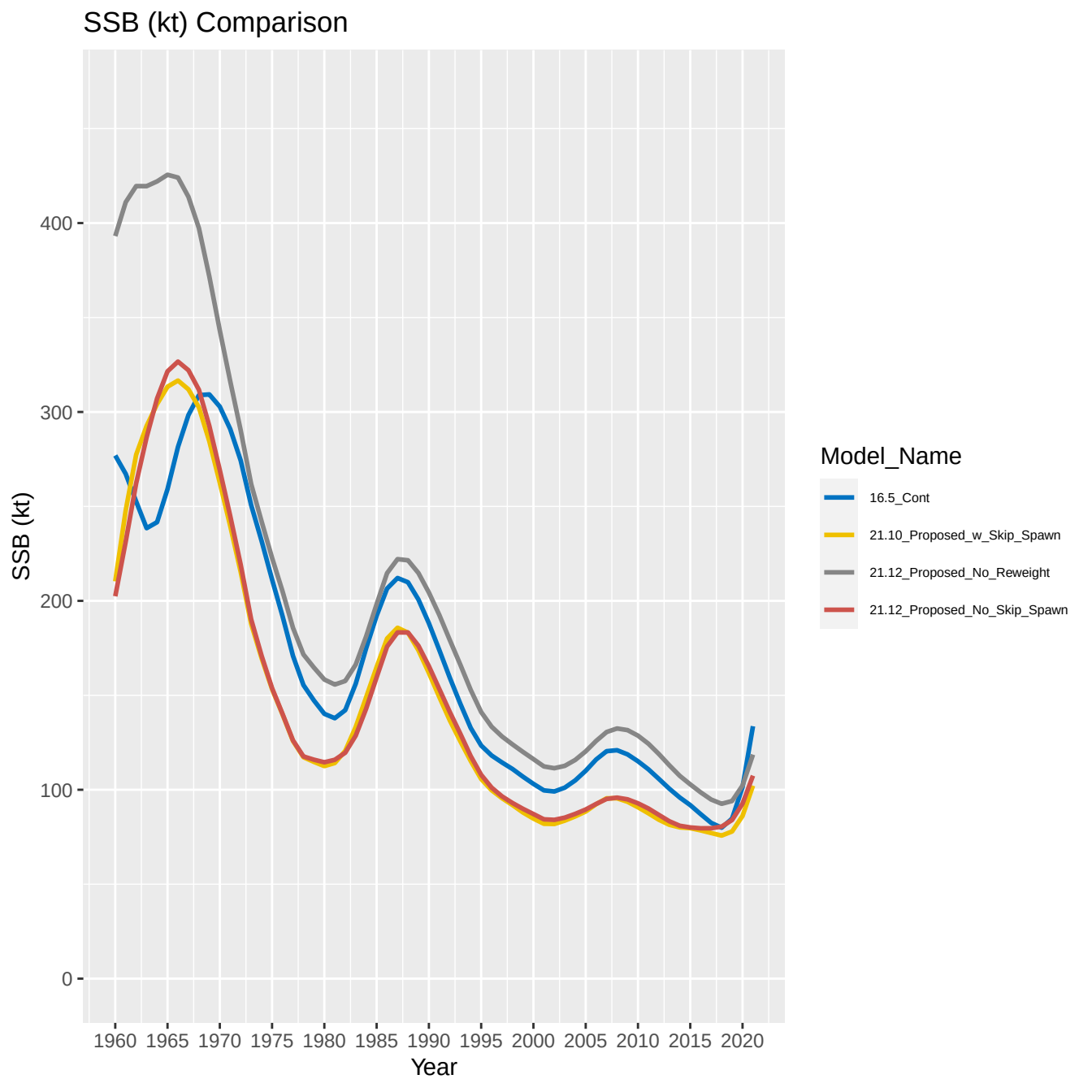
Model	Converged?	Max Grad	nLL	# Pars
21.12_Proposed_No_Reweight	TRUE	9.35527049672068e−05	1951.96	252

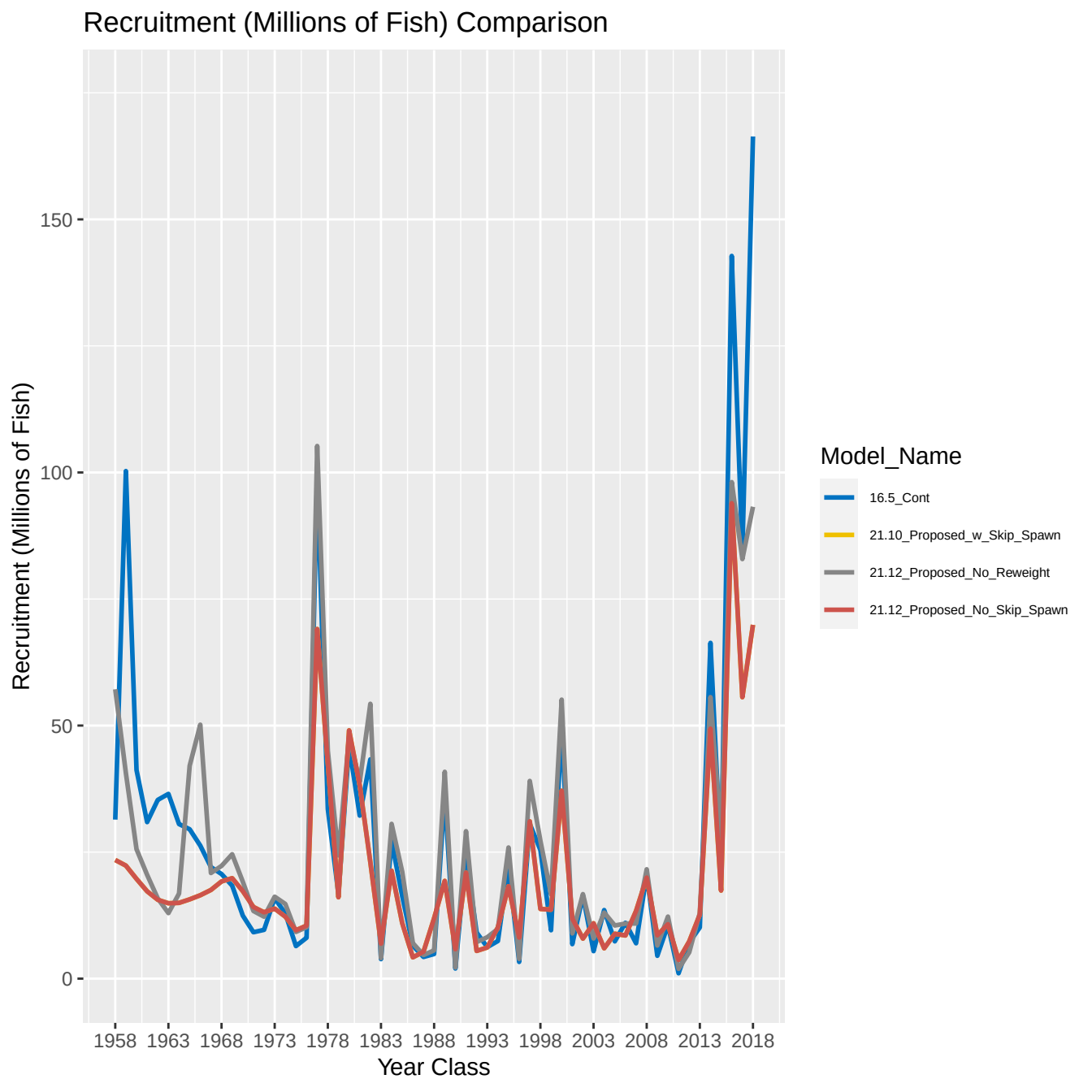
2021 SSB (kt)	SSB_40 (kt)	2021 SSB/SSB_40	Mean_Recruit	2021 F	F_40	2021 F/F_40	F_ABC	2022 ABC (kt)
118.62	120.18	0.99	25.81	0.04	0.09	0.44	0.09	48.74

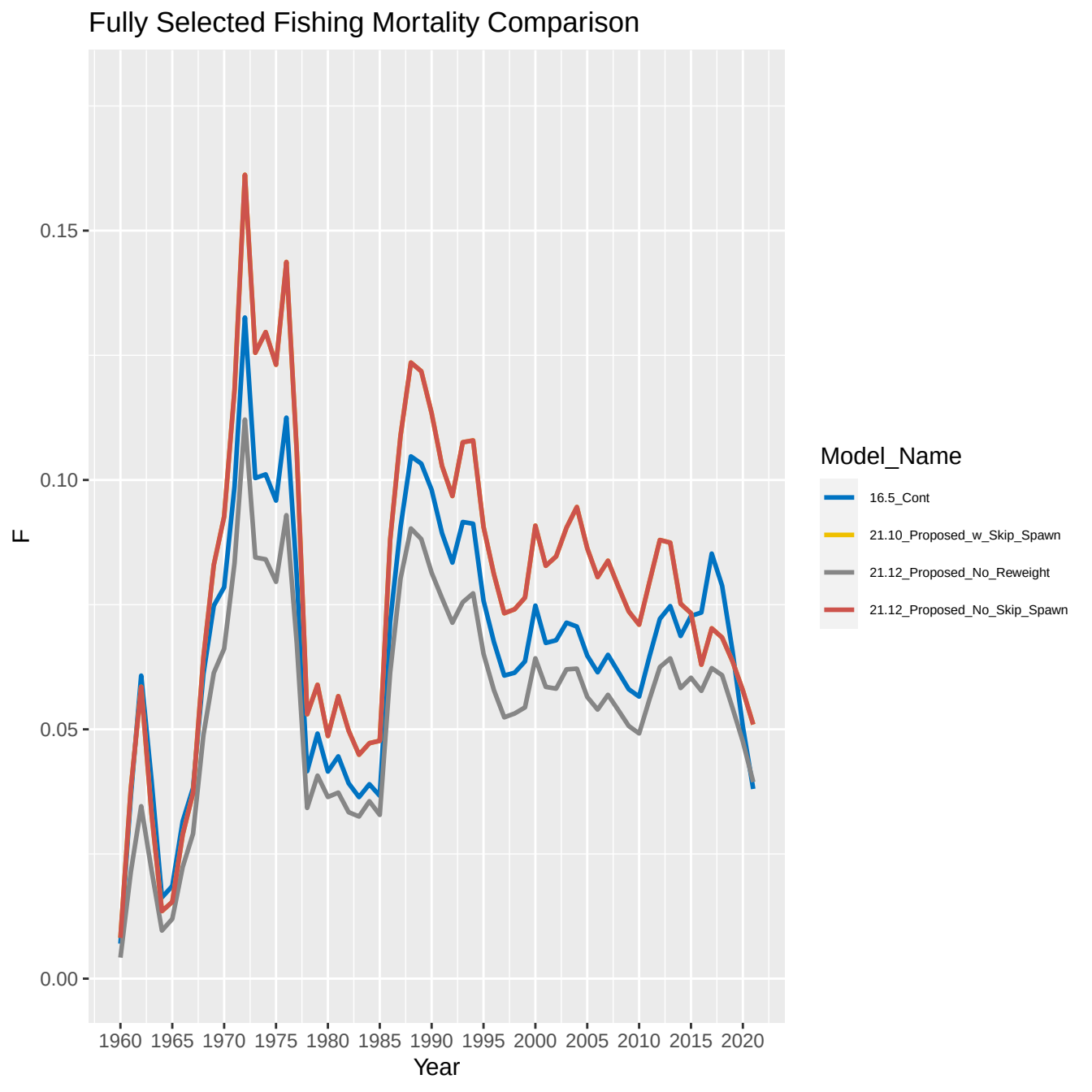
Model	Converged?	Max Grad	nLL	# Pars
21.12_Proposed_No_Skip_Spawn	TRUE	0.000678008981443245	752.92	252
16.5_Cont	TRUE	0.000205493656629296	2138.41	243
21.10_Proposed_w_Skip_Spawn	TRUE	0.000977841500663129	752.91	252
21.12_Proposed_No_Reweight	TRUE	9.35527049672068e−05	1951.96	252

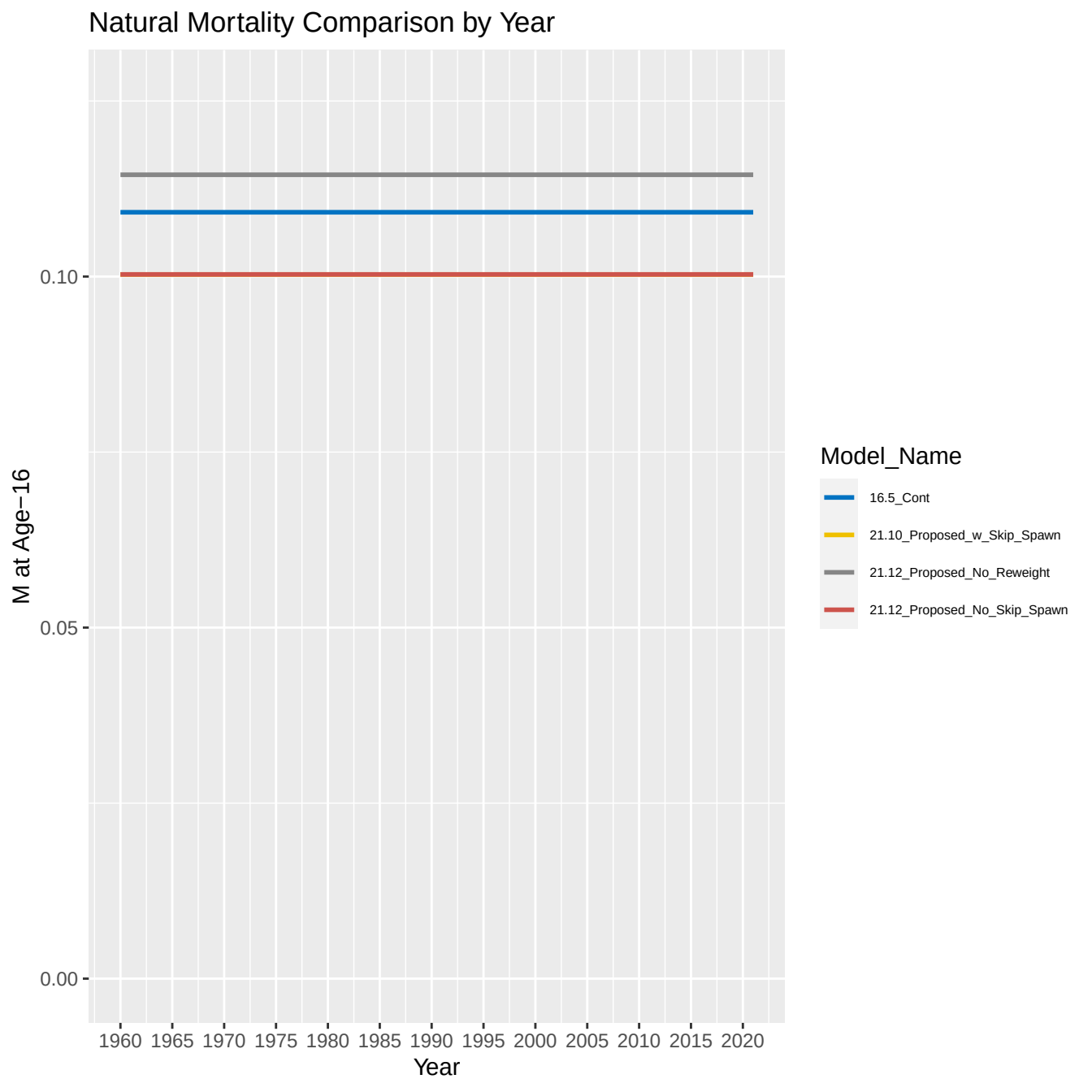
Model	2021 SSB (kt)	SSB_40 (kt)	2021 SSB/SSB_40	Mean_Recruit	2021 F	F_40	2021 F/F_40	F_ABC	2022 ABC (kt)
21.12_Proposed_No_Skip_Spawn	107.47	118.14	0.91	20.37	0.05	0.08	0.62	0.08	34.84
16.5_Cont	133.64	126.63	1.06	24.37	0.04	0.1	0.4	0.1	67.43
21.10_Proposed_w_Skip_Spawn	102.18	116.54	0.88	20.37	0.05	0.08	0.62	0.08	34.65
21.12_Proposed_No_Reweight	118.62	120.18	0.99	25.81	0.04	0.09	0.44	0.09	48.74

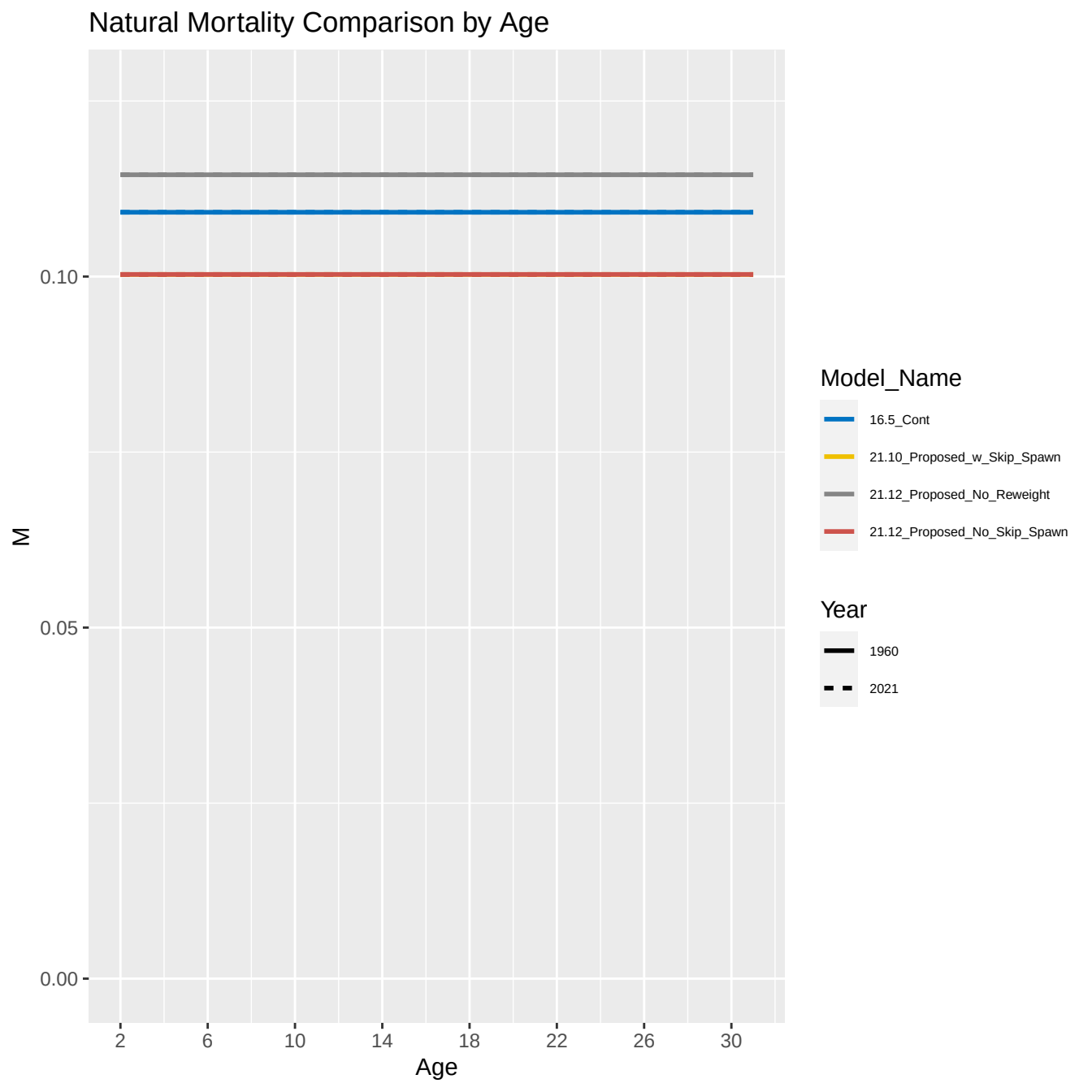

SSB (kt) Comparison
SSB (kt)
400
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
Year
B40
Model_Name
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn

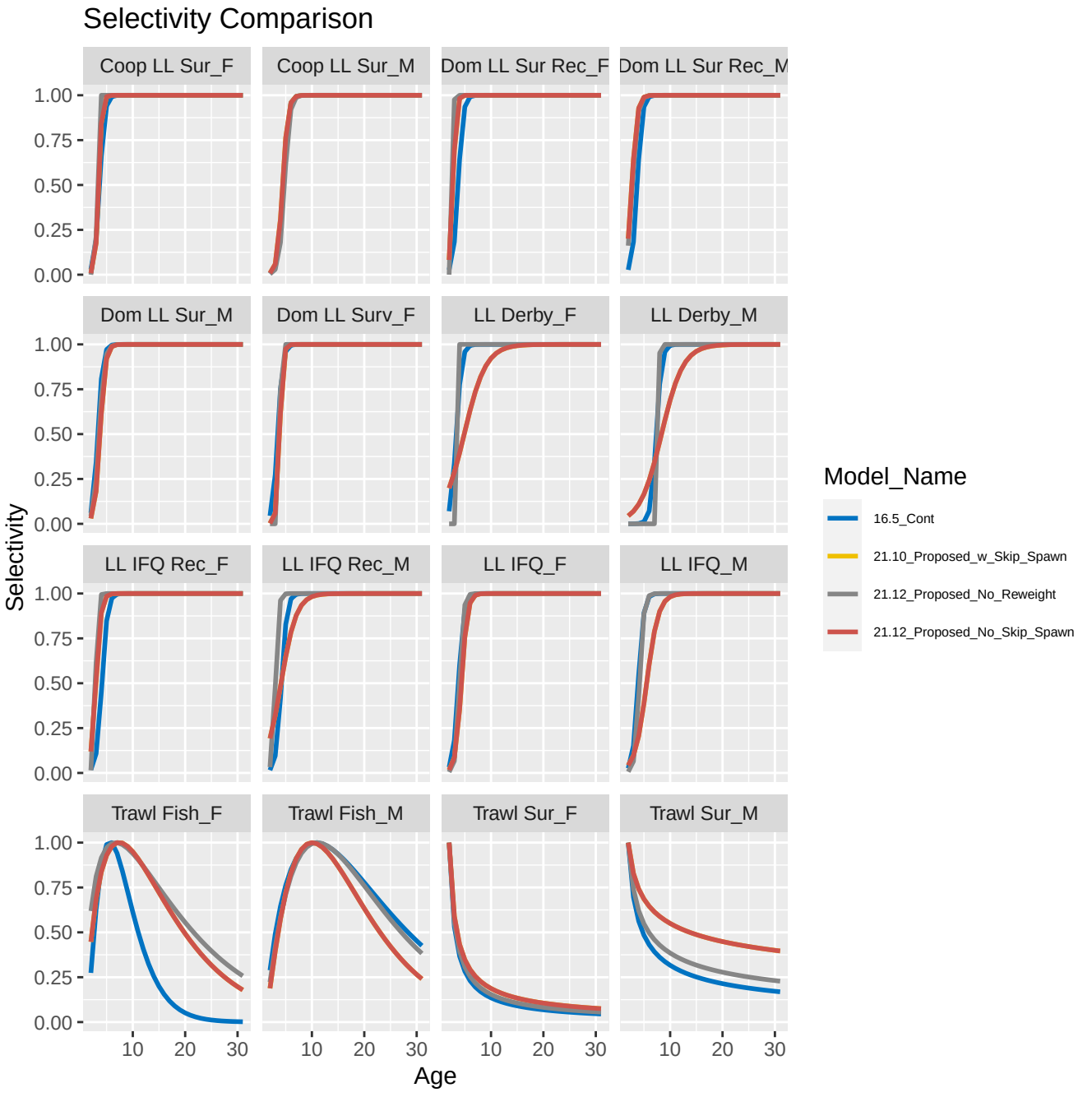

SSB (kt) Comparison
SSB (kt)
400
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
Year
Model_Name
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn

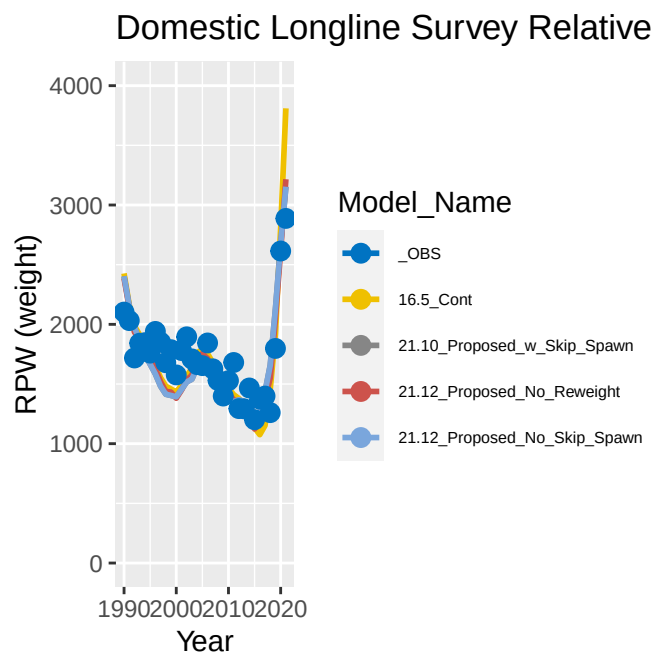

Recruitment (Millions of Fish) Comparison
Recruitment (Millions of Fish)
150
100
50
0
1958
1963
1968
1973
1978
1983
1988
1993
1998
2003
2008
2013
2018
Year Class
Model_Name
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn

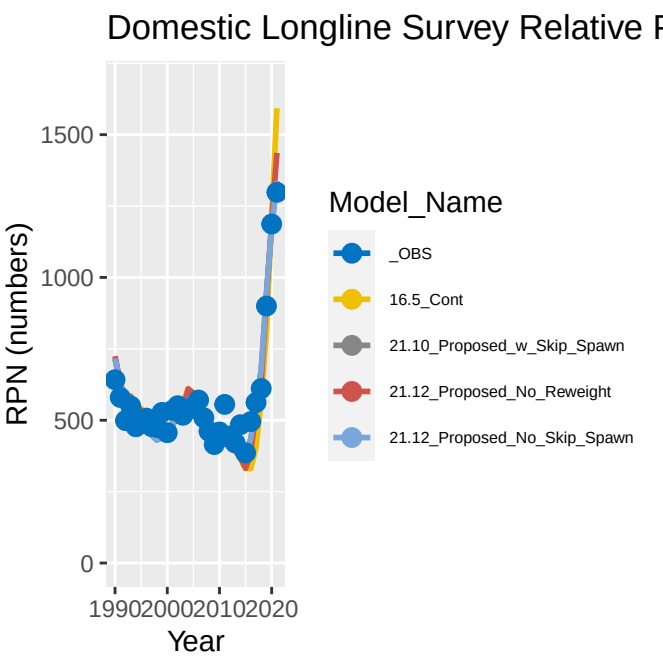

Fully Selected Fishing Mortality Comparison
F
0.15
0.10
0.05
0.00
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
Year
Model_Name
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn

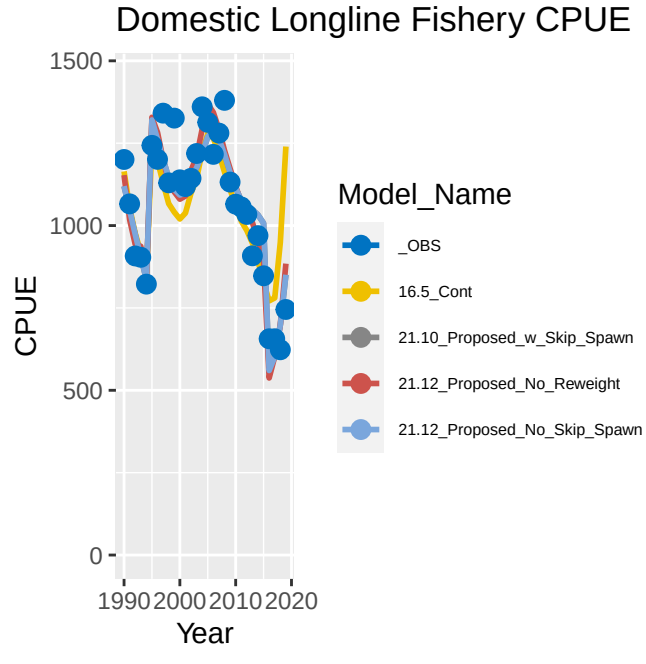

Natural Mortality Comparison by Year
M at Age−16
0.10
0.05
0.00
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
Year
Model_Name
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn

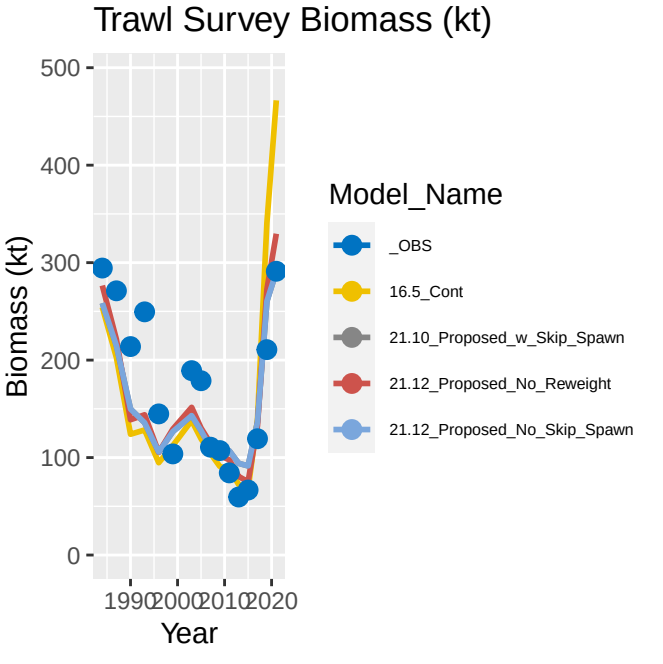

Natural Mortality Comparison by Age
0.10
0.05
0.00
M
2
6
10
14
18
22
26
30
Age
Model_Name
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn
Year
1960
2021

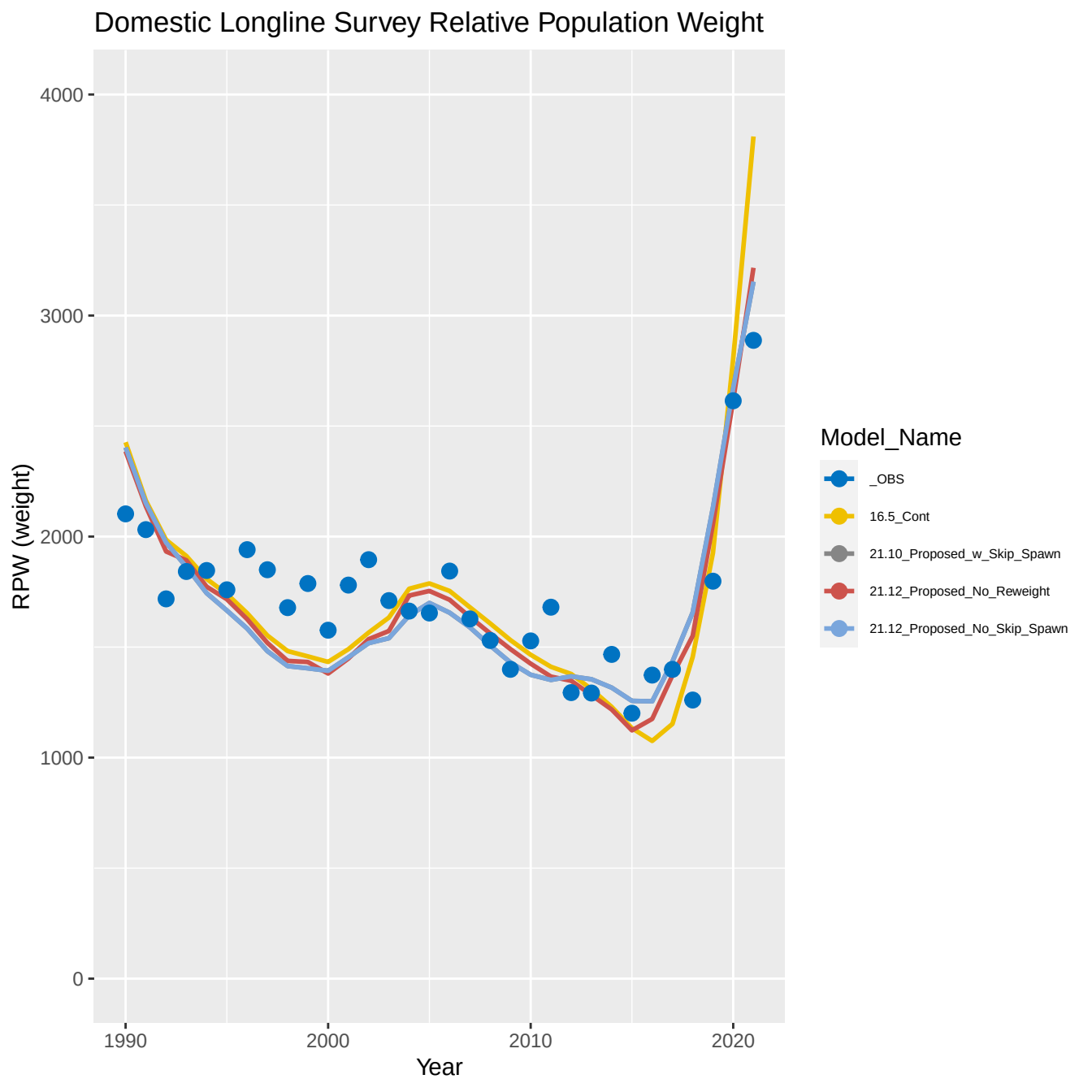

Selectivity Comparison
Coop LL Sur_F
Coop LL Sur_M
Dom LL Sur Rec_F
Dom LL Sur Rec_M
Dom LL Sur_M
Dom LL Surv_F
LL Derby_F
LL Derby_M
LL IFQ Rec_F
LL IFQ Rec_M
LL IFQ_F
LL IFQ_M
Trawl Fish_F
Trawl Fish_M
Trawl Sur_F
Trawl Sur_M
Selectivity
1.00
0.75
0.50
0.25
0.00
10
20
30
Age
Model_Name
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn

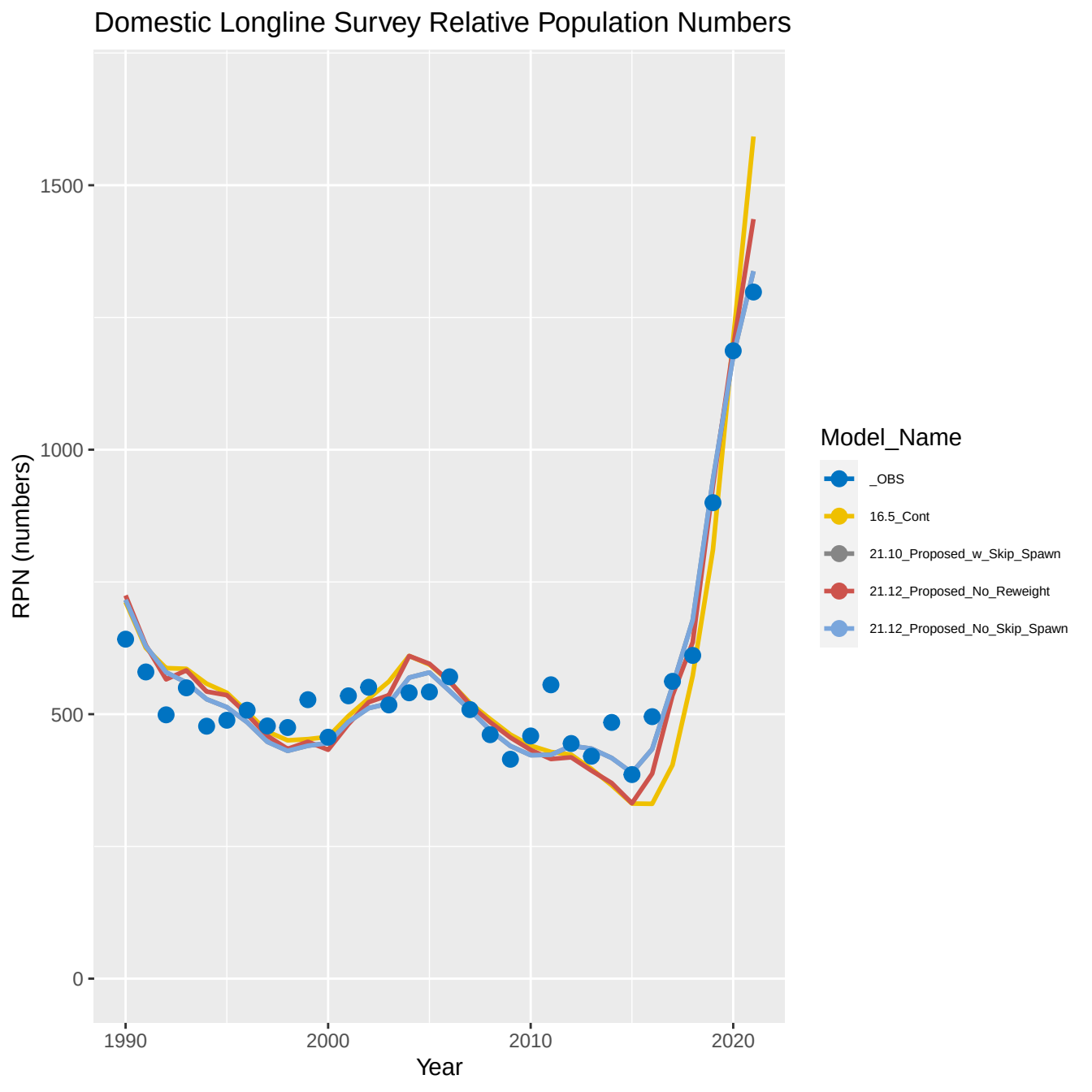

Domestic Longline Survey Relative F
RPW (weight)
4000
3000
2000
1000
0
1990
2000
2010
2020
Year
Model_Name
_OBS
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn

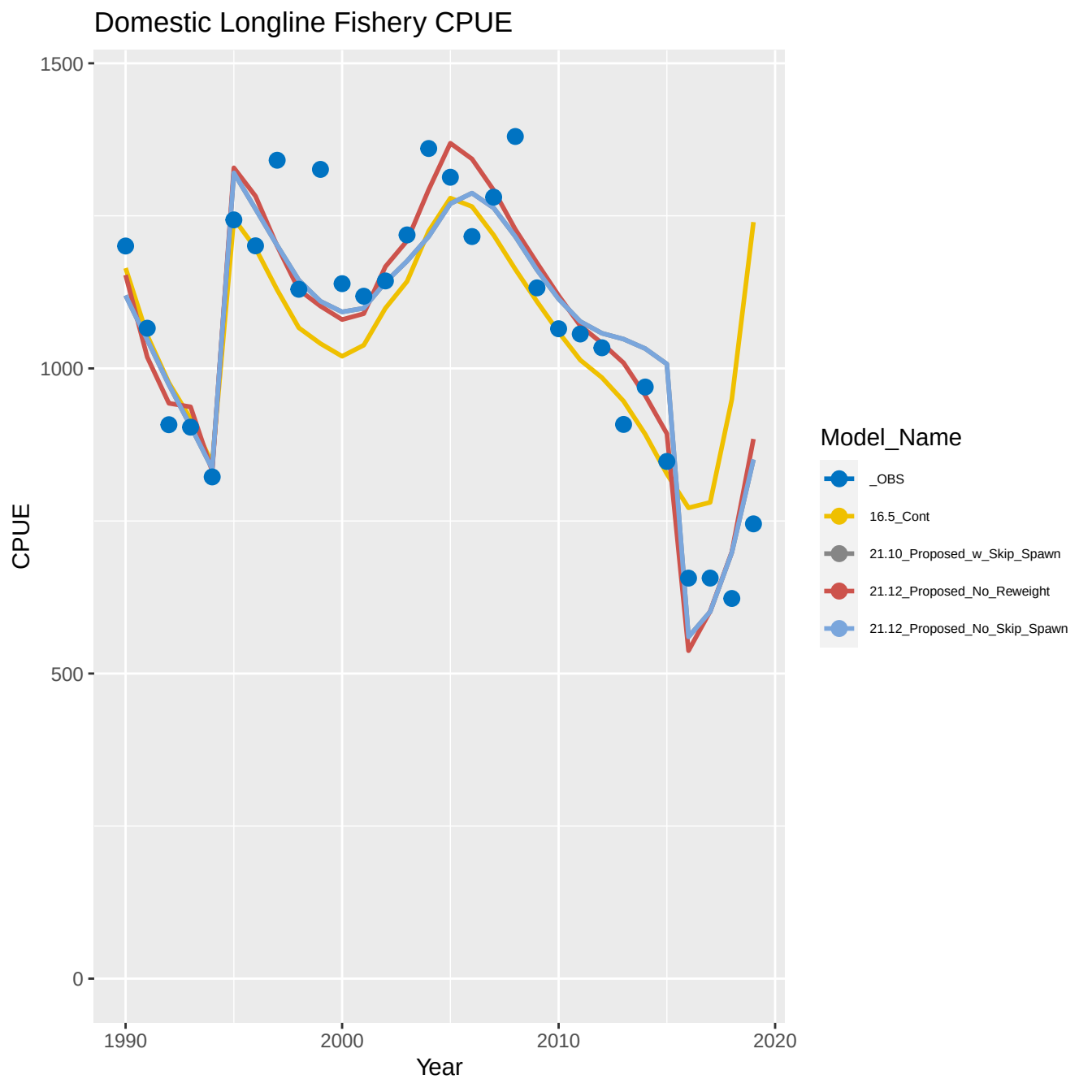

Domestic Longline Survey Relative F
RPN (numbers)
1500
1000
500
0
1990
2000
2010
2020
Year
Model_Name
_OBS
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn

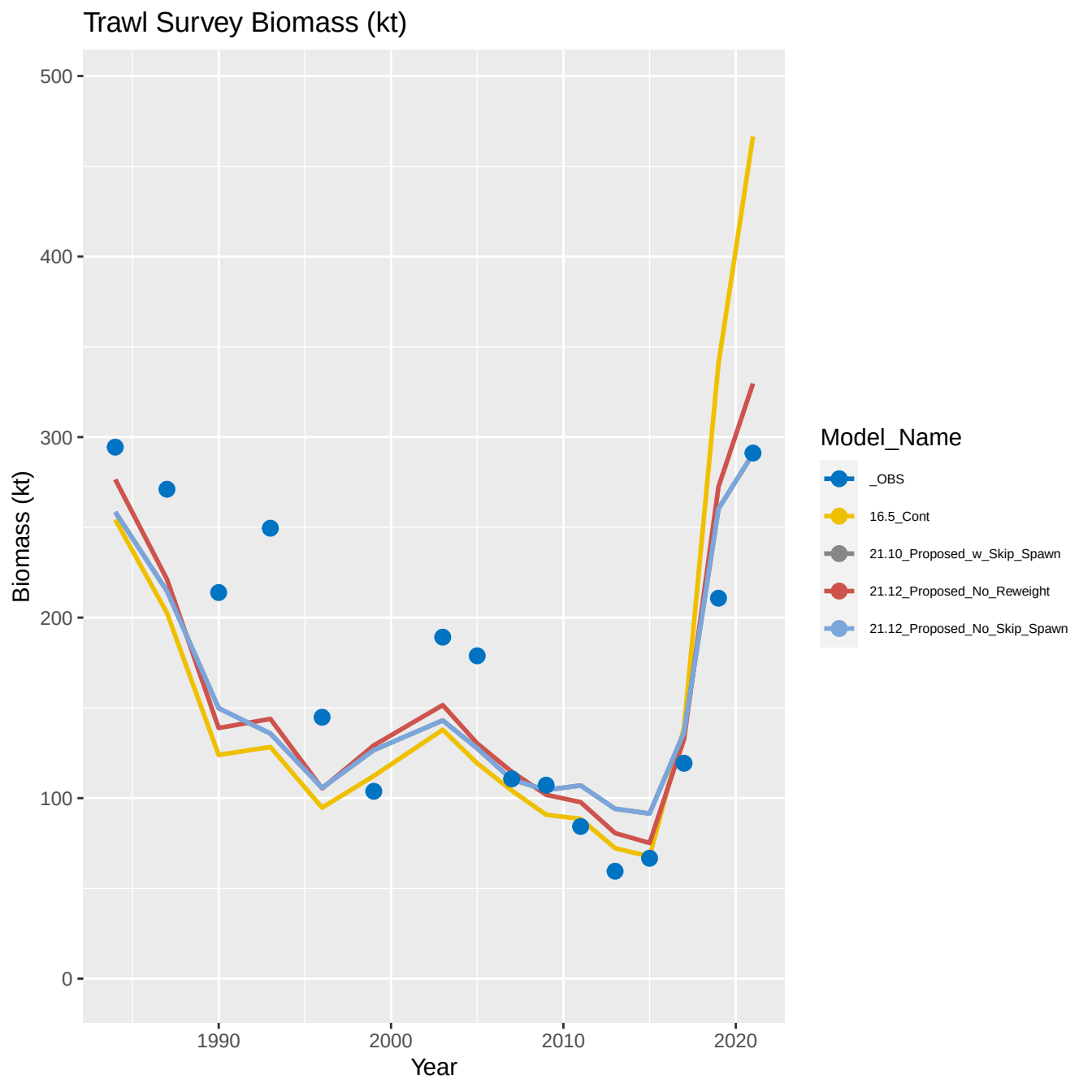

Domestic Longline Fishery CPUE
CPUE
1500
1000
500
0
1990
2000
2010
2020
Year
Model_Name
_OBS
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn


Trawl Survey Biomass (kt)
Biomass (kt)
500
400
300
200
100
0
1990
2000
2010
2020
Year
Model_Name
_OBS
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn


Domestic Longline Survey Relative Population Weight
RPW (weight)
4000
3000
2000
1000
0
1990
2000
2010
2020
Year
Model_Name
_OBS
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn


Domestic Longline Survey Relative Population Numbers
RPN (numbers)
1500
1000
500
0
1990
2000
2010
2020
Year
Model_Name
_OBS
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn


Domestic Longline Fishery CPUE
1500
1000
500
0
CPUE
1990
2000
2010
2020
Year
Model_Name
_OBS
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn


Trawl Survey Biomass (kt)
Biomass (kt)
500
400
300
200
100
0
1990
2000
2010
2020
Year
Model_Name
_OBS
16.5_Cont
21.10_Proposed_w_Skip_Spawn
21.12_Proposed_No_Reweight
21.12_Proposed_No_Skip_Spawn